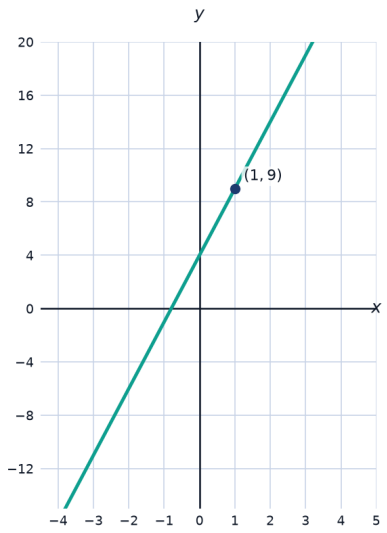
Lines & Angles

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

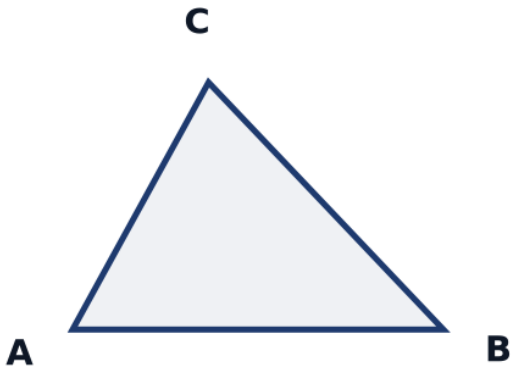
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 4$ in deductive reasoning, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

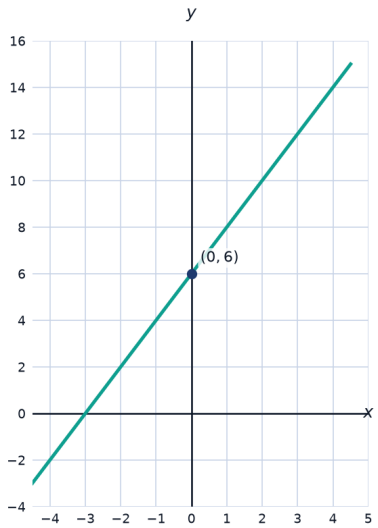
In ΔABC studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



© ClipSAT — clipsat.org | All Rights Reserved

3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 6$ in proving segment relationships, evaluate $f(2)$.

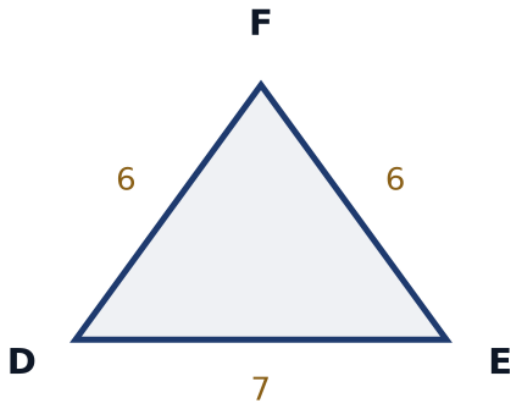


4 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for proving segment relationships consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

5 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In ΔABC studying proving angle relationships, $m\angle A = 41^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Two parallel lines L_1 and L_2 are cut by a transversal T. A pair of alternate exterior angles are given by $(7x - 14)^\circ$ and $(4x + 31)^\circ$. Find the value of x and the measure of each alternate exterior angle.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two parallel lines are intersected by a transversal line. Two consecutive interior angles have measures $(3x - 10)^\circ$ and $(2x + 25)^\circ$. Find the measure of the acute angle.

© ClipSAT — clipsat.org | All Rights Reserved

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Lines & Angles · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two angles $\angle P$ and $\angle Q$ form a linear pair. If $m\angle P = (4x + 12)^\circ$ and $m\angle Q = (2x + 18)^\circ$, find the measure of the smaller angle.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two angles $\angle 1$ and $\angle 2$ are vertical angles. If $m\angle 1 = (5x - 23)^\circ$ and $m\angle 2 = (3x + 17)^\circ$, find the measure of $m\angle 1$ and the measure of its supplement.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In ΔABC, the exterior angle at vertex C measures 134°. The two remote interior angles have measures $m\angle A = (5x - 4)^\circ$ and $m\angle B = (3x + 10)^\circ$. Find the measure of $\angle A$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In isosceles triangle ΔPQR, $PQ = PR$. The vertex angle $\angle P$ measures 40°. An angle bisector is drawn from base angle $\angle Q$ to point D on side PR. What is the measure of $\angle QDR$?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

An equilateral triangle has side lengths of 12 cm. An altitude is drawn from one vertex to the opposite side. Find the exact height and area of the triangle.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Two similar cylinders have surface areas of 64π cm^2 and 144π cm^2. If the volume of the smaller cylinder is 96π cm^3, find the volume of the larger cylinder.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Circles

Segment AB has endpoints $A(2, -1)$ and $B(6, 7)$. Find the equation of the perpendicular bisector of AB in slope-intercept form.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Area, Surface Area & Volume

In ΔPQR, ray QS bisects $\angle Q$ with S lying on side PR. If $PQ = 14$, $QR = 21$, and $PR = 25$, find the length of segment PS.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Transformations & Symmetry

Find the coordinates of the point P on directed line segment from $A(-3, 2)$ to $B(7, 12)$ that partitions the segment in the ratio $2 : 3$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Foundations & Reasoning

Triangle ΔABC has vertices $A(1, 2)$, $B(3, 2)$, and $C(1, 5)$. The triangle is dilated by a scale factor of $k = 3$ centered at the origin, and then translated by vector $\langle -2, 4 \rangle$. What are the coordinates of vertex C''?

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Coordinate Geometry

A solid metal sphere of radius 3 cm is melted down and recast into a cylinder of radius 2 cm. What is the height of the recast cylinder in cm?

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Probability

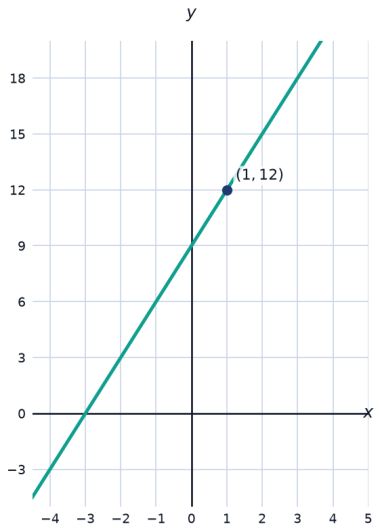
Point $A(-2, 4)$ is reflected across the line $y = x$, and the resulting point is then translated by the vector $\langle 3, -1 \rangle$. What are the final coordinates?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 9$ in conditional statements, evaluate $f(2)$.



a. 15

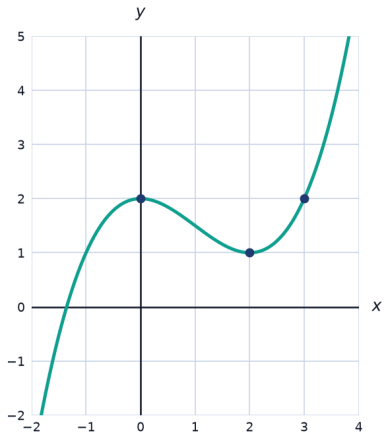
b. 18

c. 6

d. 19

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 3$ in language of logic, evaluate $f(2)$.



a. 11

b. 15

c. 8

d. 15

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In the diagram below, lines m and n are parallel. A transversal line intersects m and n. Another line segment connects a point on m to a point on n, forming a "crook-angle". Given that $\angle 1 = 120^\circ$ and $\angle 2 = 135^\circ$, what is the measure of the "crook-angle" $\angle 3$?

a. 255°

b. 105°

c. 270°

d. 75°

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two parallel lines, p and q, are intersected by a transversal line t. A pair of consecutive interior angles have measures $(4x + 9)^\circ$ and $(2x + 33)^\circ$. What is the value of x?

a. 8

b. 12

c. 23

d. 37

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two angles form a linear pair. The measures of the angles are $(3x + 10)^\circ$ and $(2x + 20)^\circ$. What is the value of x?

a. 10

b. 12

c. 30

d. 100

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Two parallel lines are cut by a transversal. One angle measures 117°. If a second angle forms a pair of alternate exterior angles with it, what is the measure of the second angle?

a. 127°

b. 63°

c. 90°

d. 117°

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two parallel lines are cut by a transversal. One angle measures 66°. If a second angle forms a pair of alternate interior angles with it, what is the measure of the second angle?

a. 66°

b. 76°

c. 114°

d. 90°

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Two parallel lines are cut by a transversal. One angle measures 55°. If a second angle forms a pair of corresponding angles with it, what is the measure of the second angle?

a. 65°

b. 90°

c. 55°

d. 125°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two parallel lines are cut by a transversal. One angle measures 144°. If a second angle forms a pair of vertical angles with it, what is the measure of the second angle?

a. 90°

b. 36°

c. 144°

d. 154°

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An angle measures 3 times its supplement. What is the measure of the angle?

a. 120°

b. 135°

c. 90°

d. 45°

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An angle measures 5 times its complement. What is the measure of the angle?

a. 65°

b. 60°

c. 75°

d. 15°

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Two parallel lines are cut by a transversal, forming a pair of corresponding angles $(6x + 1)^\circ$ and $(5x + 6)^\circ$. Find the measure of each angle.

a. 149°

b. 31°

c. 41°

d. 5°

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Alternate interior angles (parallel lines + transversal) are:

a. Supplementary

b. Complementary

c. Equal

d. Adjacent

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Two angles are vertical angles. One measures 56°. What is the measure of the other?

a. 124°

b. 56°

c. 304°

d. 34°

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two parallel lines are cut by a transversal. One angle measures 110°. Find the measure of its co-interior (consecutive interior) angle.

a. 110

b. 20

c. 80

d. 70

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two angles form a linear pair. One angle measures 161°. What is the measure of the other angle?

a. 71°

b. 19°

c. 161°

d. 199°

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two parallel lines are cut by a transversal. Two corresponding angles measure $(3x + 10)^\circ$ and $(2x + 25)^\circ$. Find the measure of the angle.

a. 15

b. 55

c. 65

d. 125

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Foundations & Reasoning

Two angles are complementary. One angle measures 7°. What is the measure of the other angle?

a. 83°

b. 97°

c. 7°

d. 173°

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Coordinate Geometry

Two parallel lines are cut by a transversal. One angle measures 29°. Find the measure of its alternate interior angle.

a. 39

b. 61

c. 29

d. 151

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Probability

Two parallel lines are cut by a transversal. One angle measures 33°. Find the measure of its alternate exterior angle.

a. 43

b. 33

c. 57

d. 147

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Triangles & Congruence

Two parallel lines are cut by a transversal. One angle measures 34°. Find the measure of its corresponding angle.

a. 34

b. 146

c. 44

d. 56

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

Two lines intersect, forming vertical angles. One angle measures 28°. Find the measure of its vertical angle.

a. 38

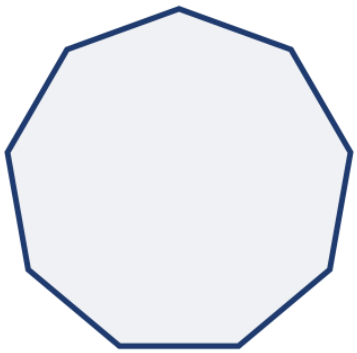
b. 152

c. 62

d. 28

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

What is the sum of the exterior angles (one at each vertex) of a convex polygon with 9 sides?



a. 1260°

b. 40°

c. 180°

d. 360°

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A quadrilateral has three angles measuring 20°, 35°, and 146°. What is the measure of the fourth angle?



a. 159°

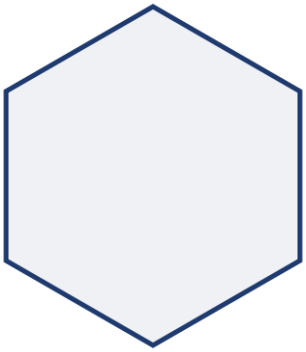
b. 169°

c. 201°

d. 20°

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Circles

How many lines of symmetry does a regular hexagon have?



a. 7

b. 6

c. 12

d. 5

© ClipSAT — clipsat.org | All Rights Reserved